Derek D. Dao

University of California, Irvine | B.S. in Computer Science and Engineering | GPA: 3.857
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Work Experience

MachineLogix (formerly AB Controls) (Irvine, CA)

Software Engineer **Oct. 2025 - Current**

- Developed 4 .NET applications that interfaced directly with automation equipment for machine control and monitoring.
- Developed 2 .NET applications that convert automation workflows into machine scripts, reducing setup time by 50%
- Created 10+ .NET hardware classes, abstracting device communication protocols into reusable, testable components.
- Developed machine scripts for robotic workflows, including liquid-handling systems, robotic arms, and bio-shakers.
- Maintained end-to-end machine control using PLC ladder logic and .NET.
- Collaborated across software and hardware teams to troubleshoot integration issues, improving automation reliability.
- Deployed an enterprise time-tracking web application with Next.js, PostgreSQL, and Docker, with role-based access, PTO management, and attendance reporting.
- Authored 4 technical documents, including user manuals and safety compliance materials for automation machines.

Science Applications International Corporation (Remote)

Full-Stack Software Engineer **June 2024 - April 2025**

- Designed and implemented RESTful APIs and built the TypeScript/React front end for a RAG LLM chatbot on AWS, integrating with a company-wide data science platform to deliver 20+ AI use cases to 3,000+ employees.
- Engineered a scalable AWS web-scraping pipeline to monitor thousands of documentation updates in real time, reducing data latency by 90% and improving LLM response freshness.
- Integrated 8 internal REST APIs connecting data pipelines to AI-assisted processes.

Backend Engineer **June 2023 - August 2023**

- Developed a Python script generating test data, automating UAT workflows and reducing engineers' testing time by 15%.
- Built and deployed a web-based internal tool (Python/Flask + D3.js) to extract and visualize PLM data, rendering hierarchical datasets as an interactive tree and improving data exploration efficiency for 50+ engineers.

Projects and Other

Kababayan — Hackathon-winning mobile app for local Filipino businesses and events **Oct. 2025**

- Built the entire backend infrastructure for local event and business data management using domain-driven services.
- Provisioned AWS infrastructure (DynamoDB, S3, IAM) using Terraform for backend deployment and data storage.
- Implemented 3 React Native screens including local event browsing, minigame, and user profile interfaces.

Envision — Hand Gesture Recognition SDK ***(UCI Dean's Choice Award Winner)*** **Sep. 2024 - March 2025**

- Developed a real-time hand recognition SDK leveraging MediaPipe and OpenCV, achieving 60+ FPS gesture detection.
- Implemented recognition interface layer for preset gestures and custom landmark-based inputs using MediaPipe outputs.
- Developed showcase applications including a gesture-based desktop manager and collaborative whiteboard using the SDK.

Fabflix — Full-Stack Movie Store **Sep. 2024 - Dec. 2024**

- Built a full-stack movie platform backed by a 500k+ record SQL database, implementing search, pagination, and cart.

Technical Skills

- **Languages:** Python, Java, JavaScript & TypeScript, C++, C#, Visual Basic, Verilog, PLC Ladder Logic
- **Cloud & Infrastructure:** AWS, Docker, Kubernetes, Terraform, Git
- **Web & Backend:** React, Next.js, Node.js, Spring Boot, REST APIs, HTML/CSS, SQL, .NET
- **AI & ML:** RAG, LLM Applications, Vector Databases, Agentic Programming, Context Engineering, MCP
- **Automation & Computer Vision:** PLC, MediaPipe, OpenCV, Embedded Systems, HMI Applications